



SARTHAK FULZELE

Full Stack AI-Developer, Independent Projects

Bachelor and Master of Technology in Mechanical Engineering, IIT Kharagpur

✉ sarthakfulzele2209@gmail.com ☎ +91-7721930950 [in linkedin.com/Sarthak-Fulzele](https://www.linkedin.com/Sarthak-Fulzele) [github/Sarthak](https://github.com/Sarthak)

EDUCATION

Bachelors of Technology, IIT Kharagpur	2021 - 2026
XII (Science), Vidya Vihar Convent High School, Chandrapur	2019 - 2021
X, BJM Carmel Academy, Chandrapur	2009 - 2019

WORK EXPERIENCE

AI-Engineer Intern | Curve-AI

Jan 2025 – Apr 2025

Python, Flask, LangGraph, Docker, AWS EC2, AssemblyAI, GPT-4o-mini, ElevenLabs, Google Calendar API, AsyncIO, Multiprocessing

Objective: Developed a production-ready voice AI assistant focused on natural conversations, automation, and scalable deployment.

- Built a real-time voice chatbot by integrating **AssemblyAI**, **GPT-4o-mini**, and **ElevenLabs** for natural, human-like dialogue
- Developed and containerized **Flask** backend APIs, deployed them on **AWS EC2**, and optimized **async I/O** with **multiprocessing**
- Implemented interruption handling and Google Calendar automation to support fluid turn-taking and seamless scheduling
- Collaborated on system architecture improvements to enhance reliability, scalability, and **low-latency** system performance

Research Internship | Deakin University | Prof. Ye Zhu and Prof. Ming Liu

Aug 2025 - Dec 2025

Python, PyTorch, LLaVA, LoRA, OpenCV, Pandas, NumPy, NLTK, Streamlit, Medical VQA, Model Evaluation Pipelines

Objective: Researched medical AI systems for diagnosis and support through fine-tuning, evaluation, and deployment pipelines.

- Researched multimodal LLMs for disease diagnosis, decision support, & scalable real-world clinical deployment use cases today
- Preprocessed 150k MedTrinity image-text pairs with **Python**, **OpenCV**, **Pandas**, **NumPy**, and **NLTK** for **LLaVA fine-tuning** pipelines
- Implemented **LoRA** and projector tuning on **LLaVA**, improving accuracy and F1-score on medical VQA benchmarks
- Built evaluation and **Streamlit** chatbot pipelines with clinician feedback, automated metrics, and interpretable clinical outputs

SKILLS

Languages:	• Golang	• Python	• C++	• Typescript	• Javascript	• SQL		
Frameworks:	• React.js	• Express.js	• Next.js	• Flask	• Django	• FastAPI		
Technologies:	• Socket.io	• MongoDB	• Git	• Kubernetes	• AWS	• Kafka	• Docker	• Postgres

PROJECTS

JAMIFY | Full-Stack Development

May 2025 – Jul 2025

ReactJs, Redux, Tailwind, NodeJs, ExpressJs, MongoDB, REST API, SocketIO

Objective: Built a real-time chat and collaborative music platform with synchronized playback and shared jam rooms for multiple users.

- Built a **real-time chat** and collaborative music platform enabling shared jam rooms and synchronized playback for multiple users
- Built secure **REST APIs** for authentication, messaging, playlist management, liked songs, and live multi-user session workflows
- Implemented **JWT-based authentication** and role-based access control to secure user accounts and broader the platform flows
- Built a responsive **React** SPA for chat, music discovery, playlists, and jam rooms with persistent global state in **Redux** Toolkit
- Implemented **Socket.IO** features for messaging, presence, typing indicators, read receipts, and synchronized playback for rooms
- Designed **synchronized controls** for play, pause, seek, track switching, and queue updates across shared **multi-user jam** sessions

PREPWISE | AI-Powered Mock Interview Platform

Jan 2026 – Mar 2026

Next.js, TypeScript, Google Gemini AI, Vapi, Firebase

Objective: Built an interview platform for candidates to practice role-specific interviews and receive structured performance feedback

- Built an AI mock interview platform using **Next.js**, **React**, and **TypeScript** to deliver role-specific practice & performance feedback
- Designed a responsive dashboard with progress cards, score snapshots, and drill-down feedback pages for each candidate
- Built reusable form flows with react-hook-form, **Zod**, and shadcn/ui to streamline authentication and validation end-to-end AI
- Engineered full interview lifecycle logic: generate set, run live voice sessions, capture transcripts, and route feedback report
- Implemented Firestore data models for accounts, interviews, and feedback, enabling per-user history, latest feeds, and retrieval
- Integrated Gemini scoring and **Vapi** voice events to evaluate communication technical depth, confidence, and role-fit in real time

BLOGGING SERVICE | Distributed Systems & Backend Engineering

Go, Kafka, Elasticsearch, Docker, Kubernetes, REST APIs

Objective: Built a scalable microservices blogging platform enabling asynchronous content publishing, real-time indexing, & fast search

- Built a **Go microservices** blogging platform with **Kafka & Elasticsearch**, supporting **concurrent** publishing and low-latency search
- Designed an async Kafka ingestion pipeline to decouple blog writes from indexing, to improve API responsiveness under load
- Built a Kafka consumer service that processed blog events & indexed documents into Elasticsearch, keeping it stable under load
- Implemented a search API with keyword filters and relevance-ranked retrieval across blog posts for low-latency content discovery
- Containerized services with **Docker** & deployed them on **Kubernetes** with **scalable** pods, service discovery, and safe **rollout flows**
- Collaborated on service contracts, error handling, and observability to keep the platform stable for production-grade operations

COMPETITIONS

[OPEN-SOFT GENERAL CHAMPIONSHIP](#) | **Movie Website** | **SILVER** Mar 2024
MongoDB, Express.js, React.js, Node.js, JavaScript, Hugging Face, All MiniLM L6 V2, REST APIs, NLP
Objective: Built an AI-powered movie discovery platform focused on semantic search, recommendations, and admin operations.

- Built a MERN movie platform with semantic search, responsive UI, and real-time query suggestions for smooth content discovery
- Utilized Hugging Face's All MiniLM L6 V2 API to improve search efficiency by leveraging NLP techniques to deliver relevant results
- Designed a recommendation engine plus admin panel for user control, moderation workflows, analytics, and content management

CONFERENCE

[BACHELORS THESIS PROJECT 1](#) | **IIT Kharagpur** | Prof. V Mahendra Reddy Aug 2024 - Dec 2024
A Chemical Kinetic Study on the Influence of Micro-Mix of Hydrogen (H2) & Hydrogen Peroxide (H2O2) in Ammonia Mixtures.
Ansys Chemkin, Python, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn, Regression Analysis, ANN
Objective: Investigated ammonia combustion stability using Ansys Chemkin simulations & ANN modeling for emission analysis

- Generated a large dataset of **30,000+ Ansys Chemkin simulations** on **NH3-H2O2 mixtures** for flame stability & emission analysis
- Applied regression analysis and heatmap visualizations to map effects of preheat temperature, equivalence ratio, and NOx outputs
- Trained an ANN using ReLU and Adam to deliver accurate predictions with strong generalization across unseen test datasets
- Got the **Youngest presenter title** at the Net-zero Emissions Technology (**NOET-2024**) **Conference** held at **IIT (ISM) Dhanbad**

COURSE PROJECTS

[BACHELOR'S THESIS PROJECT 2](#) | **IIT Kharagpur** | Prof. V. Mahendra Reddy Jan 2025 - Apr 2025
Comparative Analysis of Machine Learning & Deep Learning Models for Predicting Ignition Delay Times of Pure & Blended Fuels
Ansys Chemkin, Python, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn, Regression Analysis, ANN

- Conducted **Chemkin simulations** with **400k+ data points** on five individual fuels & ten blends across various operating conditions
- Implemented LR, DT, RF, XGBoost, and ANN models; ANN achieved <3% error in predicting blend interactions
- Improved model robustness using Bayesian tuning, stratified cross-validation, and physics-informed features
- Derived practical insights for **fuel design and emission control**; research is currently in final stages of **peer-reviewed publication**

[MASTER'S THESIS PROJECT 1](#) | **IIT Kharagpur** | Prof. Sankha Deb Aug 2025 - Dec 2025
AI-Enhanced Neural Network Methodology for Process Planning of Rotationally Symmetrical Parts in Flexible Manufacturing
Python, Neural Networks, CAPP, Manufacturing Automation, Feature Encoding, Industry 4.0

- Engineered AI-driven CAPP for rotational parts, encoding manufacturing heuristics into neural networks for smart planning system
- Built feature encoding and data pipelines mapping tolerances and surface finish to predict machining sequences accurately scales
- Optimized flexibility to recommend machining sequences, outperforming rigid rule-based systems in adaptability & upkeep gains
- Validated on rotational parts, showing strong generalization and lower manual engineering effort for Industry 4.0 adoption gains

POSITIONS OF RESPONSIBILITY

[CORE ORGANISING TEAM MEMBER](#) | **Spring Fest IIT Kharagpur** Aug 2022 - Jun 2023

- Led sponsorship campaigns across Mumbai and Pune, securing strong funding growth and surpassing the previous fest edition
- Expanded the fest through International and Indie Carnivals, attracting artists and performers from over fifteen countries
- Co-led a team of 46 team members, extending the outreach to 850+ colleges & ensuring coverage through 250+ published articles
- Conducted Bhubaneswar region prelims, registering 900+ participants & recording 150% growth in registrations over prior edition

[SECRETARY- SOCIAL AND CULTURAL](#) | **Pandit Madan Mohan Malaviya Hall of Residence IIT Kharagpur** Aug 2022 - Apr 2023

- Organized & managed cultural events that boosted community engagement, resident participation, and vibrant hall life year-round
- Planned Talent Hunt, cultural shows, and inter-hall contests, ensuring smooth execution, strong turnout, & memorable experiences
- Coordinated with student members, external vendors to manage logistics, including budgeting, marketing, and event setup

EXTRA CURRICULAR ACTIVITIES

- Secured District Rank 5 and Maharashtra Rank 123 in Junior level General Science Examination conducted by ICAD
- Participated in Inter-Hall General Championship Nukkad Naatak and achieved Gold among 23 halls, showcasing theatrical talent
- Participated in Illumination 2022, making contribution in the hall's collective efforts and achieving Silver among 23 halls
- Obtained level 6 certification in Harmonium proficiency and achieved Excellent Grade at NCA, IIT Kharagpur
